

INTEGRATING **YOUTUBE CHANNEL** ----- **DATABASES**

Analyze Engagement, Growth, and Trends



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DATA SOURCES

We are using two publicly available YouTube datasets sourced from Kaggle to build our YouTube Channel Performance Database.

Dataset 1: Top Subscribed YouTube Channels

Contains:

- Rank
- Channel name
- Subscriber count
- Total video views
- Video count
- Category
- Year started

Dataset 2: YouTuber Dataset

Contains:

- Channel name
- Subscribers
- Average views
- Average likes
- Average comments
- Category
- Country
- Content type

Reason for Choosing These Datasets

Both datasets were selected from Kaggle because they provide reliable, diverse, and publicly available information on YouTube channels. Each dataset offers different attributes, allowing us to merge them into a connected database and explore meaningful business questions around growth, engagement, and content trends.

About

BACKGROUND OF DATA

YouTube is one of the largest digital platforms globally, with millions of creators who vary by category, country, engagement level, and subscriber base.

Businesses use YouTube analytics to:

- Partner with creators
- Run advertisements
- Identify fast-growing categories
- Predict channel growth
- Understand audience behavior

Our two datasets give a 360-degree view of YouTube channels subscriber size, engagement metrics, performance, and content style.



PROJECT OBJECTIVES

Data Preparation & Cleaning

- Standardized column names and formats for consistency
- Removed duplicates and handled missing values
- Converted data types (numeric, text, factors) for accurate processing
- Created a unified Channel_ID to merge datasets reliably

Database Integration & Relationship Building

- Designed three relational tables: Channels, Ranking_Subscribers, Engagement_Metrics
- Established primary and foreign key relationships for clean joins
- Normalized the structure to reduce redundancy and maintain data integrity
- Built a scalable schema supporting future datasets

Analysis & Insights Using SQL Queries

- Applied SQL logic to examine trends in engagement, ranking, country, and category performance
- Calculated metrics such as engagement ratios, category averages, and influencer potential
- Generated insights to identify high-value channels for marketing and strategic decisions
- Used SQL-style queries in R (dplyr) to automate analysis workflows

DATABASE DESIGN

1. Channels Table

- Channel_ID (Primary Key)
- Channel_Name, Category, Country, Year_Started, Content_Type

2. Ranking Table

- Channel_ID (Foreign Key)
- Rank, Subscribers, Total_Views, Video_Count

3. Engagement Table

- Channel_ID (Foreign Key)
- Avg_Views, Avg_Likes, Avg_Comments

BUSINESS QUESTION

How do subscriber numbers relate to engagement metrics such as views, likes, and comments?

Objective:

Analyze how subscriber count relates to engagement (views, likes, comments) to measure true audience activity.

What We Did:

- Joined channel, ranking, and engagement tables
- Calculated engagement efficiency metrics: View Rate, Like Rate, Comment Rate
- Compared whether large channels actually maintain strong engagement levels

SQL Query Used:

```
SELECT c.Channel_Name,
       r.Subscribers,
       e.Avg_Views,
       e.Avg_Likes,
       e.Avg_Comments,
       (e.Avg_Views / r.Subscribers) AS View_Rate,
       (e.Avg_Likes / r.Subscribers) AS Like_Rate,
       (e.Avg_Comments / r.Subscribers) AS Comment_Rate
FROM Channels c
JOIN Ranking_Subscribers r USING (Channel_ID)
JOIN Engagement_Metrics e USING (Channel_ID);
```

SQL Query 1 Output – Subscriber vs Engagement Analysis

youtube_channel	subscribers_x	average_views	average_likes	average_comments	view_rate
T-Series	234000000.0	135200.0	5600.0	223.0	0.0005777777777777778
Cocomelon - Nursery Rhymes	152000000.0	5100000.0	57100.0	0.0	0.03355263157894737
SET India	150000000.0	27900.0	996.0	7.0	0.000186
MrBeast	128000000.0	104000000.0	4000000.0	74000.0	0.8125
PewDiePie	111000000.0	1700000.0	109200.0	3100.0	0.015315315315315315

BUSINESS QUESTION

Which YouTube content categories perform best in terms of subscriber growth and audience engagement?

Objective:

Identify which YouTube categories (Music, Gaming, Education, etc.) achieve the highest growth and engagement.

What We Did:

- Grouped data by category
- Calculated average subscribers, views, likes, and comments
- Ranked categories by engagement strength

SQL Query Used:

```
SELECT c.Category,
       AVG(r.Subscribers) AS Avg_Subscribers,
       AVG(e.Avg_Views) AS Avg_Views,
       AVG(e.Avg_Likes) AS Avg_Likes,
       AVG(e.Avg_Comments) AS Avg_Comments
FROM Channels c
JOIN Ranking_Subscribers r USING (Channel_ID)
JOIN Engagement_Metrics e USING (Channel_ID)
GROUP BY c.Category
ORDER BY Avg_Views DESC;
```

SQL Query 2 Output – Category Performance

category_x	subscribers_x	average_views	average_likes	average_comments
Autos & Vehicles	18900000.0	1700000.0	30100.0	63.0
Comedy	18841666.666666668	2573347.9166666665	108603.70833333333	2768.0416666666665
Education	25691891.891891893	497624.3243243243	9799.81081081081	705.7567567567568
Entertainment	20643093.922651935	1150500.223756906	42070.872928176796	1134.1491712707182
Film & Animation	24000000.0	578986.4864864865	20876.837837837837	528.5405405405405

SQL Query Used:

```
SELECT c.Country,  
       COUNT(*) AS Total_Channels,  
       AVG(r.Subscribers) AS Avg_Subscribers,  
       AVG(e.Avg_Views) AS Avg_Views,  
       AVG(e.Avg_Likes) AS Avg_Likes,  
       AVG(e.Avg_Comments) AS Avg_Comments  
FROM Channels c  
JOIN Ranking_Subscribers r USING (Channel_ID)  
JOIN Engagement_Metrics e USING (Channel_ID)  
GROUP BY c.Country  
ORDER BY Avg_Views DESC;
```

BUSINESS QUESTION

How does a channel's country of origin influence its subscriber growth and engagement performance?

Objective:

Compare how channel performance varies across different countries.

What We Did:

- Grouped channels by country
- Calculated average subscribers, engagement metrics, and channel counts
- Identified high-engagement countries vs. high-volume countries

SQL Query 3 Output – Country Performance

country	subscribers_x	average_views	average_likes	average_comments
Algeria	17600000.0	191000.0	10200.0	579.0
Argentina	16500000.0	7991100.0	286487.1666666667	11446.666666666666
Bangladesh	16433333.333333334	82700.0	1615.6666666666667	66.33333333333333
Brazil	19017543.85964912	348054.3859649123	25414.052631578947	4634.912280701755
Colombia	21050000.0	385750.0	15540.0	563.4

SQL Query Used:

```
SELECT c.Channel_Name,  
       r.Rank,  
       r.Subscribers,  
       e.Avg_Views,  
       (e.Avg_Views / r.Subscribers) AS  
Engagement_Ratio  
FROM Channels c  
JOIN Ranking_Subscribers r USING (Channel_ID)  
JOIN Engagement_Metrics e USING (Channel_ID)  
ORDER BY Engagement_Ratio DESC;
```

BUSINESS QUESTION

Do subscriber-based rankings accurately reflect actual engagement levels across channels?

Objective:

Determine if top-ranked channels also show high engagement, or if subscriber rankings are misleading.

What We Did:

- Joined ranking and engagement data
- Calculated Engagement Ratio (Views per Subscriber)
- Highlighted channels with high rank but low engagement

SQL Query 4 Output – Ranking vs Engagement Consistency

youtube_channel	rank_x	subscribers_x	average_views	engagement_ratio
Bizarrap	409	18200000	44500000.0	2.4450549450549453
MrBeast	5	128000000	104000000.0	0.8125
Sidemen	445	17600000	11600000.0	0.6590909090909091
Village Cooking Channel	359	19500000	11400000.0	0.5846153846153846
Marta and Rustam	588	15100000	8600000.0	0.5695364238410596

BUSINESS QUESTION

Which creators demonstrate the strongest engagement quality and are best suited for brand collaborations?

Objective:

Identify creators with high-quality engagement—ideal for brand partnerships.

What We Did:

- Evaluated Like-to-View and Comment-to-View ratios
- Ranked creators based on engagement quality
- Highlighted channels with highly responsive audiences

SQL Query Used:

```
SELECT c.Channel_Name,
       r.Subscribers,
       e.Avg_Views,
       e.Avg_Likes,
       e.Avg_Comments,
       (e.Avg_Likes / e.Avg_Views) AS Like_View_Ratio,
       (e.Avg_Comments / e.Avg_Views) AS Comment_View_Ratio
FROM Channels c
JOIN Ranking_Subscribers r USING (Channel_ID)
JOIN Engagement_Metrics e USING (Channel_ID)
ORDER BY Like_View_Ratio DESC, Comment_View_Ratio DESC;
```

SQL Query 5 Output – Top Creators for Brand Collaborations

youtube_channel	subscribers_x	like_view_ratio	comment_view_ratio
Bispo Bruno Leonardo	14400000	0.3423048168322119	0.2509879312186265
Pastor Antônio Júnior	12000000	0.2289156626506024	0.02072289156626506
Piuzinho	12500000	0.21621621621621623	0.010333863275039745
BANGTANTV	73100000	0.1875908166230747	0.004359197907585004
Lana Del Rey	12900000	0.1829787234042553	0.011955420466058764

BUSINESS QUESTION

How can engagement efficiency be used to identify high-value influencers for marketing and partnership strategies?

Objective:

Use engagement efficiency to find high-ROI creators for marketing and sponsorship decisions.

What We Did:

- Calculated Engagement Efficiency (Views ÷ Subscribers)
- Filtered for channels with >30% engagement efficiency
- Generated a shortlist of creators ideal for campaigns

SQL Query Used:

```
SELECT c.Channel_Name,
       c.Category,
       c.Country,
       r.Subscribers,
       e.Avg_Views,
       (e.Avg_Views / r.Subscribers) AS Engagement_Efficiency
FROM Channels c
JOIN Ranking_Subscribers r USING (Channel_ID)
JOIN Engagement_Metrics e USING (Channel_ID)
WHERE (e.Avg_Views / r.Subscribers) > 0.30
ORDER BY Engagement_Efficiency DESC;
```

SQL Query 6 Output – Influencer Strategy (Efficiency > 30%)

youtube_channel	category_x	subscribers_x	average_views	efficiency
Bizarrap	Comedy	18200000	44500000.0	2.4450549450549453
MrBeast	Entertainment	128000000	104000000.0	0.8125
Sidemen	Entertainment	17600000	11600000.0	0.6590909090909091
Village Cooking Channel	Howto & Style	19500000	11400000.0	0.5846153846153846
Marta and Rustam	Comedy	15100000	8600000.0	0.5695364238410596

Business Question Insites

1

Big channels don't always have strong engagement; smaller channels often perform better relative to their size.

2

Music has scale, but categories like Education and Gaming deliver stronger engagement quality.

3

USA & India lead in channel volume, while the Philippines and Mexico show stronger engagement efficiency.

4

High ranking doesn't guarantee high engagement; mid-ranked channels can outperform top-ranked ones.

5

Creators with high like/view ratios offer the best ROI for brand collaborations.

6

Creators with engagement efficiency above 30% are the most valuable for targeted partnerships.

MANAGERIAL IMPLICATION

1

For YouTube

- Enhance ranking algorithms by incorporating engagement-based metrics (views, likes, comments) instead of relying heavily on subscriber count.
- Improve category and channel discovery by highlighting creators with consistently strong engagement efficiency.
- Use insights from category and geography trends to refine recommendation systems and regional content promotion.
- Strengthen platform analytics dashboards by integrating engagement ratios to support creators and advertisers.

2

For Marketers & Brands

- Identify high-ROI influencers using engagement quality indicators such as Like/View and Comment/View ratios.
- Prioritize partnerships with creators who show strong engagement efficiency rather than just large subscriber numbers.
- Choose content categories strategically—Education, Gaming, Music, and Lifestyle show different engagement advantages depending on campaign goals.
- Use country-level performance insights to plan region-specific influencer campaigns and market expansions.

3

For Content Creators

- Benchmark performance against category and country averages to understand competitive positioning.
- Focus on improving audience interaction (comments, likes, feedback loops) to increase engagement quality.
- Use engagement gaps to adjust content strategy—topics, posting frequency, video types, or audience targeting.
- Leverage insights to attract brand collaborations by showcasing strong engagement metrics, not just follower count.

Conclusion

- This project successfully integrated two separate YouTube datasets into a unified, relational database that provides a complete view of channel performance.
- Through rigorous data cleaning, normalization, and relationship design, we built a reliable structure that supports deeper analytics beyond basic subscriber counts.
- SQL-driven analysis revealed important insights on engagement efficiency, category trends, geographic performance, and creator influence—highlighting that true impact comes from engagement quality, not just audience size.
- The results offer actionable guidance for marketers, brands, creators, and the YouTube platform by identifying high-value categories, countries, and creators for strategic decisions.
- Overall, this system demonstrates how thoughtful database design combined with SQL logic can transform raw YouTube data into meaningful, business-ready insights—laying the foundation for scalable and future-ready analytics.



THANK YOU